



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CCT) (CBCS) Degree Examinations

May-June 2025

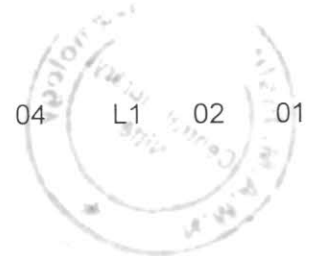
22CCT202 – CONSTRUCTION & CONTRACT MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define ethics and Discuss any three ethical principles.	04	L*1	01	01
b)	Discuss the following with relevant equations. i) Power Sizing method of cost estimating. ii) Cost Indexing.	06	L2	01	02
c)	Prepare a preliminary estimate for large Hydro Electric Power Plant, with following data: i) Capacity augmentation: 1000 MW, cost per MW is Rs 1.5 Cr. ii) Environmental mitigation cost is 2% of total project cost. iii) Resettlement cost is 5% of the project cost. iv) Project all risks and contingencies is 2% of capacity augmentation cost. v) Project management fee is 2% of gross. vi) Design, survey, geotechnical etc – 5% of capacity augmentation cost. vii) Escalation cost is 7% of gross.	10	L3	01	03
2. a)	Discuss the phases of lifecycle costing and specify their benefits.	04	L2	01	02
b)	Explain the following. i) Empirical costing methods. ii) Bottom-Up Estimating.	06	L2	01	02
c)	Prepare a budgetary estimate for the following project details and determine the cost per square area. Type of construction - Residential building of Ground + 5 floors. Area of each floor - 350 sqm. Number of units - 4 Basic construction cost of civil works is Rs 30000/sqm, Cost of Electrification is 8% of gross project cost, Cost of Plumbing & Sanitary arrangement is 4% of gross cost, Cost of Architectural works is 2% of gross cost, Cost of interior works is 10% of gross cost and Cost of all other works is 5% of gross cost. Total indirect cost is 6% of gross, Profit is 5% of gross and Taxes are 4% of gross.	10	L3	01	03
Unit – II					
3. a)	Discuss any four factors affecting labor output or production efficiency.	04	L2	02	02
b)	Discuss four common 'Cost Estimator Practice Traits'.	06	L2	02	02
c)	For a highway project, the required quantity of earthwork is one lakh cubic meter. The job is to be finished in 40 days. To perform this job, it is decided to use an excavator working for 10 hours in a day. If the output of each excavator is 80 cum per hour and the crew consists of 1 Highly Skilled Labor (HSL), 1 Semi Skilled Labor (SSL) and 2 Unskilled Labors (UL), paid at Rs 30,000/month for HSL, Rs 25,000 for SSL and Rs 20,000 for UL all inclusive. Compute : i) Total number of excavators ii) Labor production rate iii) Unit labor cost for excavation	10	L3	02	03



4. a) What is Bid rigging? List out at least five signs of possible bid rigging.
 b) Highlight the impact of following cost aspects on inventory cost of a project.
 i) Purchase costs
 ii) Order cost
 iii) Holding costs
 iv) Unavailable cost
 c) Timber formwork for a roof slab of a single storey building needs to be completed in 3 days of 8 hours working each. Total qty of work is 600 sqm. Labor crew consisting of 2 skilled workers and 1 unskilled worker is employed does fair face form work of 1.5 sqm per team hour.
 Estimate:
 (i) Total team hours required
 (ii) Man hours involved
 (iii) Number of teams or crew gang required
 (iv) Total work force employed
 (v) Unit rate of work, by considering skilled worker is paid Rs 900 per day and unskilled at Rs 700 per day
 (vi) Hourly labor rate for this activity.

06 L2 02 04

10 L3 02 03

Unit – III

5. a) Write short notes on following
 i) Selective tendering ii) Negotiated tender
 b) What Is Construction Bidding? Highlight the activities involved in tender processing.
 c) Write short notes on following referring to bidding models.
 i) Individual rate loading
 ii) Front-end loading. iii) Back end loading
 6. a) State four essential requirements of valid contract.
 b) i) What is bid capacity? Determine the bid capacity for the following project details.
 Project duration – 4 years
 Maximum work value – 500 Crore
 Existing commitments at present value – Rs 200 Crore
 c) For a balanced bid, the bid amount is Rs 63.71 lakhs Following BOQ constitute the bid items.

04 L1 03 01

06 L1 03 01

10 L1 03 01

04 L2 03 01

06 L2 03 03

Sl No	Item	UOM	Quantity	Final rate	Amount
1	Earth work excavation	Cum	200	180	36000
2	RCC works	Cum	300	17500	52,50,000
3	Masonry work	Cum	300	3500	10,50,000
4	DPC	Sqm	100	350	35000
				Rs(Lakhs)	63.71

By detailed investigation, it was found that quantity of RCC work will decrease from 300 to 50 Cum, whereas quantity of masonry work will increase from 300 Cum to 3000 Cum.

Determine the final unit price for both RCC work and masonry work using quantity error exploitation method. The bid price is unchanged.

10 L3 03 03

Unit – IV

7. a) Explain with reference to FIDIC, the following
 i) Liquidated Damages and Bonus for Early Completion
 ii) Breach of Contract
 b) As per Indian Contract Act 1872, Define "CONTRACT", and discuss any four construction contract provisions to be negotiated with owner.

04 L1 04 01

06 L2 04 02

c)	Discuss in brief the following clauses of a standard set of conditions of contract				
	i) Performance security/ security deposit				
	ii) Suspension of work				
	iii) Time limit for completion	10	L2	04	01
8.	a) Discuss the following types of contracts.				
	i) Cost Plus Percentage Contract.				
	ii) Cost Plus Fixed Fee Contract.	04	L2	04	02
	b) Explain with reference to FIDIC, the following				
	i) Liquidated Damages and Bonus for Early Completion				
	ii) Breach of Contract	06	L2	04	01
	c) The estimated construction cost for building is 9.25 Cr and the cost variation is +/- 11.5%. Compute total actual cost to owner and profit or loss to him, if owner offers following type of contracts.				
	(i) Cost plus percentage fee (CPPF) at 6.5% agreed profit.				
	(ii) Cost plus Fixed Fee (CPFF) at 6.5% profit on estimated cost,				
	(iii) Fixed price Contract (FPC) @ 6.5% profit on estimate.	10	L3	04	03
Unit – V					
9.	a) Discuss the four things to do when faced with a dispute in contract management.	04	L2	05	02
	b) List out six situations leading to "Contractors Claims on Owner".	06	L1	05	01
	c) i) Discuss "BREACH OF CONTRACT AND MATERIALITY OF BREACH" in project management.				
	ii) Explain the following frequent breach of contract situations in civil contracts				
	- The Spearin Doctrine				
	- Misrepresentation	10	L2	05	02
10.	a) Discuss the situations leading for extra work and change order, without dispute in Civil Engineering Contracts.	04	L2	05	02
	b) Discuss briefly importance of "COMMUNICATION" in Good Contract Management"	06	L2	05	02
	c) Discuss the following frequent breach of contract situations.				
	i) Failure to make payment for completed work				
	ii) Interference with contractual performance				
	iii) Non disclosure of superior knowledge				
	iv) Improper termination of contract	10	L2	05	02

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
